Team Gradient

AI Loan Eligibility Assessment Assistant

- Multi-Agent Workflow Orchestration using RAG and LangGraph
- Finance Domain — UST Enterprise Capstone

Problem Statement

Operational Inefficiency

Manual loan processing is slow and inefficient, leading to high operational costs.

Decision Inconsistency

Risk evaluation is inconsistent across different manual assessors.

Compliance Bottlenecks

Manual policy validation increases delays and risk of human error.

Strategic Need

Need for AI-driven workflow automation to modernize lending practices.

Goal: Automate loan eligibility assessment

Project Objectives

Assessment Automation

Automate loan eligibility assessment to increase speed and reduce operational overhead.

Agent Orchestration

Build a robust multi-agent orchestration workflow for specialized task handling.

Intelligent Routing

Implement router-based approvals to ensure logic-driven decision paths.

Policy Retrieval (RAG)

Integrate RAG for precise policy retrieval to maintain strict compliance standards.

Transparency

Improve workflow transparency with clear logging and state tracking.

Human-in-the-loop

Support human approval checkpoints for critical edge cases and risk mitigation.

Why Multi-Agent Architecture?

Specialized Agents

Individual agents dedicated to handling distinct and specific tasks within the ecosystem.

Better Modularity

Deconstructs complex processes into manageable, interchangeable modules.

Simple Maintenance

Streamlines orchestration and simplifies long-term maintenance of the system.

Enterprise Simulation

Mimics complex enterprise workflows for more realistic and robust simulations.

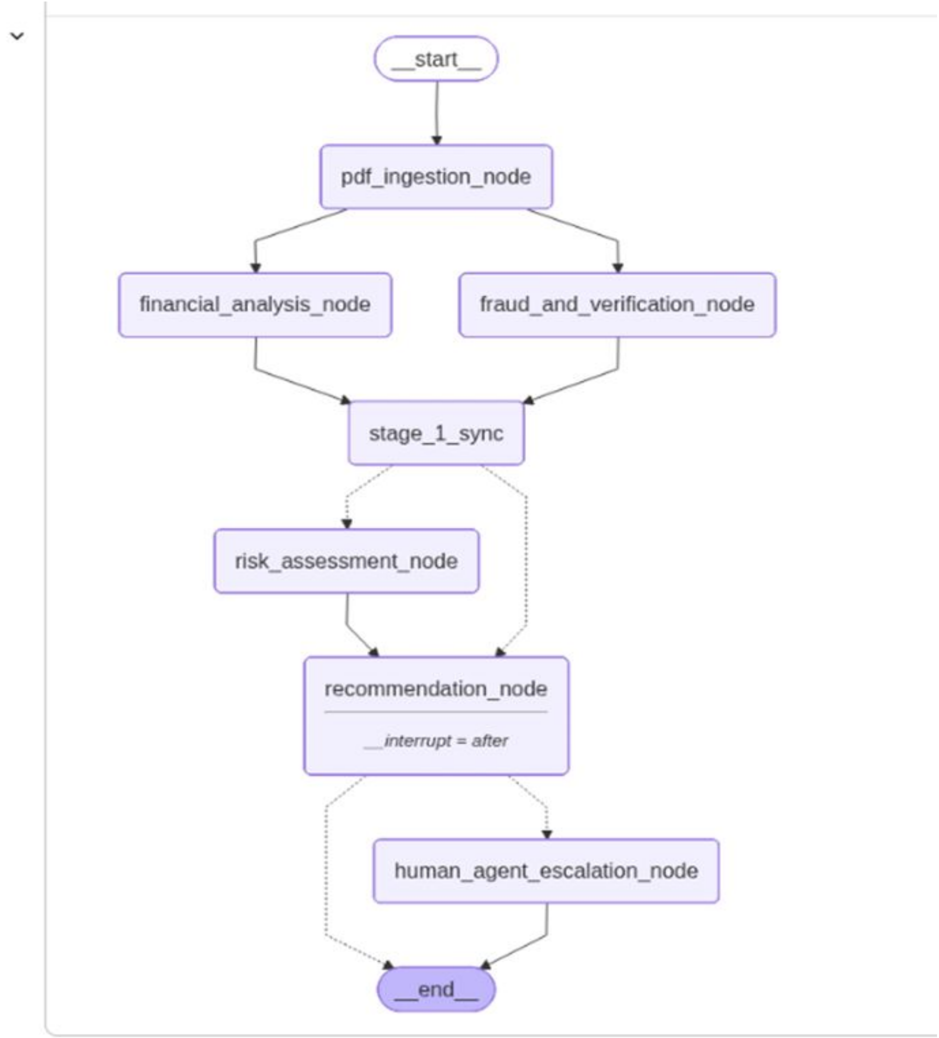
Scalable & Flexible

Effortlessly scales to meet growing demands while maintaining operational flexibility.

Agent Responsibilities

Agent	Responsibility
Financial Analysis Agent	Analyze applicant financial profile
Fraud & Verification Agent	Verify applicant documents and fraud signals
Risk Assessment Agent	Classify LOW / MEDIUM / HIGH risk
Recommendation Agent	Generate loan eligibility recommendation
Human Escalation	Review risky applications manually

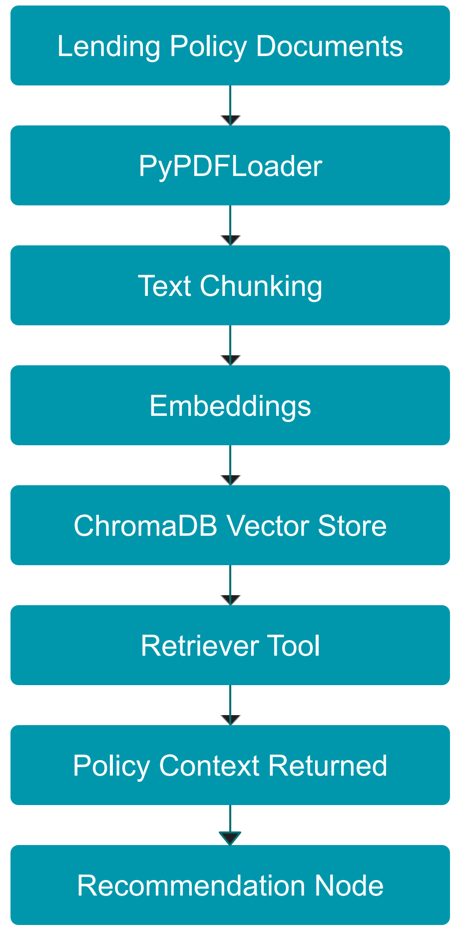
System Architecture



Technology Stack

Category	Technology Used
Agent Framework	LangGraph + LangChain
LLM	Gemini 2.5 Flash
RAG Components	LangChain (PyPDF Loader, Recursive Character TextSplitter)
Vector Database	ChromaDB
Programming Language	Python

RAG Pipeline



Live Demo

Step 1	Upload Loan Application PDF
Step 2	Extract Applicant Information
Step 3	Execute Multi-Agent Workflow
Step 4	Retrieve Lending Policies
Step 5	Generate Risk Assessment
Step 6	Produce Recommendation
Step 7	Human Escalation for High-Risk Cases

Workflow Design

01. PDF Ingestion

Automated intake and parsing of loan application documents.

02. Parallel Analysis

Concurrent financial analysis and fraud verification processing.

03. Synchronization Stage

Merging parallel data streams into a unified workflow state.

04. Risk Assessment

Evaluation of applicant data against lending policies.

05. Recommendation

Generation of automated loan approval or denial suggestions.

06. Human Escalation

Manual review by loan officers for high-risk or complex cases.

Memory & Context Handling

Workflow state is maintained across LangGraph nodes.

State includes:

- Applicant details
- Financial analysis results
- Fraud verification status
- Risk classification
- Recommendation output

Benefits:

Shared workflow state Unified visibility across all processing agents.

Stateful orchestration Seamless handoffs between complex decision nodes.

Node synchronization Ensures data integrity during parallel execution.

Error Handling & Robustness

Input Integrity

Validation checks for invalid inputs to ensure data quality from the start.

Graceful Failures

Smart handling for missing fields, preventing system crashes during processing.

API Resilience

Automated retry and fallback mechanisms for external API dependencies.

Stateful Recovery

Persistent state tracking allows for seamless recovery during workflow interruptions.

Challenges Faced

Multi-Agent State

Handling complex multi-agent state transitions effectively.

RAG Latency

Optimizing retrieval latency for processing large PDF documents.

Risk Classification

Ensuring consistency in AI-driven automated risk classification.

Data Integration

Integrating diverse financial data formats and complex schemas.

Future Enhancements

OCR Support

Support for scanned PDFs to broaden data ingestion capabilities.

Banking Integration

Real-time connectivity with financial institutions for instant verification.

Explainable AI

Enhanced transparency in AI decision-making for regulatory compliance.

Fraud Detection

Advanced heuristics to identify and mitigate sophisticated fraud attempts.

Cloud Deployment

Full migration to cloud-native architecture for infinite scalability.

Monitoring

Enhanced real-time dashboard for comprehensive system health tracking.

Conclusion

The project successfully demonstrates a scalable architecture for:

Multi-agent orchestration

RAG pipelines

Router-based workflows

AI-driven decision systems

Automated enterprise loan assessment

Q&A

Questions & Discussion

We'd love to hear your thoughts and inquiries.

Thank You